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Subject: Java
Assignment: 5
Form Number: 251101107

Classes, Objects and Constructors

1. Define a class of type Student that has rollno, name and age as private data members. Define SetData() and GetData() as public member functions with appropriate functionality. Write a program that declares 2 student objects, initializes the first at run-time and second by reading from console, and then displays both student's data.

```java
import java.util.*;
class Student{
    private int rno;
    private String name;
    private int age;

void setStudent(int r,String n,int a){
    rno = r;
    name = n;
    age = a;
}

void getStudent(){
    System.out.println(rno+" "+name+" "+age);
}
public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    Student s1 = new Student();
    s1.setStudent(12,"Nakul",23);
    System.out.println("Student 1 details are: ");
    s1.getStudent();

    Student s2 = new Student();

    System.out.println("Enter roll no");
    int r = sc.nextInt();

    System.out.println("Enter name");
    String n = sc.next();

    System.out.println("Enter age");
    int a =  sc.nextInt();

    s2.setStudent(r,n,a);

    System.out.println("Student 2 details are: ");
    s2.getStudent();

}
```

```
C:\Users\DAC-User32\Desktop\nakul>javac Student.java

C:\Users\DAC-User32\Desktop\nakul>java Student
Student 1 details are:
12 Nakul 23
Enter roll no
22
Enter name
om
Enter age
20
Student 2 details are:
22 om 20

C:\Users\DAC-User32\Desktop\nakul>
```

2. Create a class Person with attributes name, age and country. Implement methods to set and get these attributes. Create an object of this class, set its attributes, and print out the details.

```java
class Person{
    String name;
    int age;
    String country;

void setPerson(String n,int a,String c){
    name=n;
    age = a;
    country = c;
}

void getPerson(){
    System.out.println(name+" "+age+" "+country);
}
public static void main(String[] args) {
    Person p1 = new Person();
    System.out.print("Person Details are: ");
    p1.setPerson("Nakul",21,"Pakistan");

    p1.getPerson();
}
}
```

```
C:\Users\DAC-User32\Desktop\nakul>javac Person.java

C:\Users\DAC-User32\Desktop\nakul>java Person
Person Details are: Nakul 21 Pakistan

C:\Users\DAC-User32\Desktop\nakul>
```

3. Constructor Overloading: Extend the Person class from the previous problem and add multiple constructors (default, parameterized, etc.) to initialize the attributes. Also, include a method to display the details.

```java
import java.util.*;

class BankAccount{
    int accountno;
    String account_type;
    double balance1;
    double amount;
    static double interest_rate = 5.52;

   void setBankAccount(int a ,String at,double b){
       accountno = a;
       account_type = at;
       balance1 = b;
    }
    void deposit(double amount){
       balance1 +=amount;

    }
    void withdrawfund(double amount){
       balance1 -=amount;
    }
    void display(){
       System.out.println("account number:"+accountno);
       System.out.println("account type:"+account_type);
       System.out.println(" balance"+balance1);

    }

    public static void main(String[] args) {
       BankAccount b1 = new BankAccount();
       BankAccount b2 = new BankAccount();

       b1.setBankAccount(123,"saving",100);
       b2.setBankAccount(1234,"current",200);

       b1.display();
       System.out.println("interest_rate"+interest_rate);
       System.out.println();

       b1.deposit(300);
       b1.display();
       System.out.println();

       b1.withdrawfund(100);
       b1.display();
       System.out.println();


       }
    }
```

```
C:\Users\DAC-User32\Desktop\nakul>javac BankAccount.java

C:\Users\DAC-User32\Desktop\nakul>java BankAccount
account number:123
account type:saving
 balance100.0
interest_rate5.52

account number:123
account type:saving
 balance400.0

account number:123
account type:saving
 balance300.0


C:\Users\DAC-User32\Desktop\nakul>
```

4. Using this: Modify the Person class to include a method that displays the name and age of the object. Use this keyword to differentiate between class variables and method parameters. Implement a method to compare two Person objects based on their age.

```java
import java.util.*;
class Person1{
     String name;
     int age;


void setPerson(String n,int a){
    name=n;
    age = a;

}

void display(){
    System.out.println(name+" "+age+" ");
}
void comparing(Person1 p){
    if(age>p.age){
        System.out.println(name+" is big");
    }else if (age==p.age){
        System.out.println(" Both are same age ");
    }else{
        System.out.println(p.name+" is big: ");
    }
}
public static void main(String[] args) {
    Scanner s = new Scanner(System.in);
    Person1 p1 = new Person1();
    Person1 p2 = new Person1();

    System.out.println("Enter NAME:");
    String n = s.next();

    System.out.println("Enter AGE:");
    int a = s.nextInt();

    p1.setPerson(n,a);

    System.out.println("Enter NAME:");
    String n2 = s.next();

    System.out.println("Enter AGE:");
    int a2 = s.nextInt();

    p2.setPerson(n2,a2);

    p1.display();
    p2.display();

    p1.comparing(p2);
}
}
```

```
C:\Users\DAC-User32\Desktop\nakul>javac Person1.java

C:\Users\DAC-User32\Desktop\nakul>java Person1
Enter NAME:
NAKUL
Enter AGE:
21
Enter NAME:
OM
Enter AGE:
20
NAKUL 21
OM 20
NAKUL is big

C:\Users\DAC-User32\Desktop\nakul>
```

5. Static Variable: Create a class BankAccount with accno, accType, Balance and static variable interestRate. Initialize it using a static block. Implement methods to deposit and withdraw funds. Create objects and display details.

```java
import java.util.*;

class BankAccount{
    int accountno;
    String account_type;
    double balance1;
    double amount;
    static double interest_rate = 5.52;

   void setBankAccount(int a ,String at,double b){
        accountno = a;
        account_type = at;
        balance1 = b;
    }
    void deposit(double amount){
        balance1 +=amount;

    }
    void withdrawfund(double amount){
        balance1 -=amount;
    }
    void display(){
        System.out.println("account number:"+accountno);
        System.out.println("account type:"+account_type);
        System.out.println(" balance"+balance1);

    }

    public static void main(String[] args) {
        BankAccount b1 = new BankAccount();
        BankAccount b2 = new BankAccount();

        b1.setBankAccount(123,"saving",100);
        b2.setBankAccount(1234,"current",200);

        b1.display();
        System.out.println("interest_rate"+interest_rate);
        System.out.println();

        b1.deposit(300);
        b1.display();
        System.out.println();

        b1.withdrawfund(100);
        b1.display();
        System.out.println();



    }
}
```

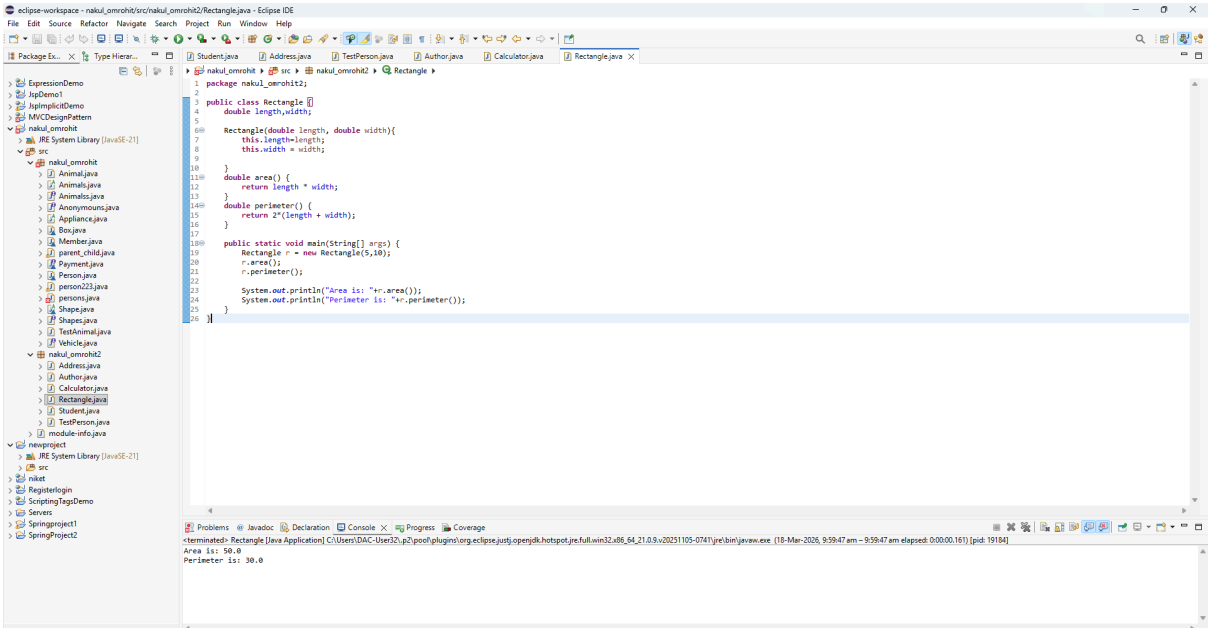
```
C:\Windows\System32\cmd.e

C:\Users\DAC-User32\Desktop\nakul>javac BankAccount.java

C:\Users\DAC-User32\Desktop\nakul>java BankAccount
account number:123
account type:saving
 balance100.0
interest_rate5.52

account number:123
account type:saving
 balance400.0

account number:123
account type:saving
 balance300.0


C:\Users\DAC-User32\Desktop\nakul>
```

6. Static Method: Add a static method to the BankAccount class from the previous problem to calculate interest based on a given balance and interest rate. Also, implement a method to display the account details including balance and interest earned.

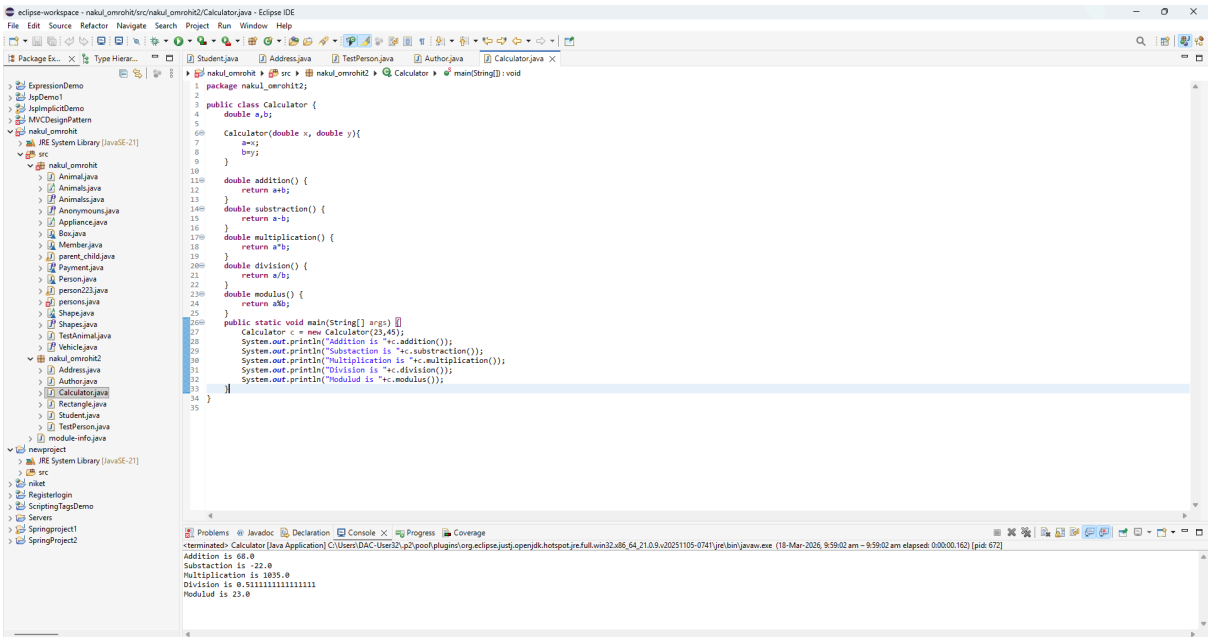
```java
class Bank{
    int accountno;
    String account_type;
    double balance1;
    double amount;
    static double interest_rate = 5.52;

   void setBank(int a ,String at,double b){
        accountno = a;
        account_type = at;
        balance1 = b;
    }
    void deposit(double amount){
        balance1 +=amount;

    }
    void withdrawfund(double amount){
        balance1 -=amount;
    }
    static double interestrate(double balance1){
        return balance1 * interest_rate/100;
    }

    void display(){
        double interest = interestrate(balance1);
        System.out.println("account number:"+accountno);
        System.out.println("account type:"+account_type);
        System.out.println("balance"+balance1);
        System.out.println("Interest Earned "+interest);

    }

    public static void main(String[] args) {
        Bank b1 = new Bank();
        Bank b2 = new Bank();

        System.out.println("Interest rate is: "+interest_rate);
        System.out.println();


        b1.setBank(123,"saving",100);
        b2.setBank(1234,"current",200);

        b1.display();
        System.out.println();

        b1.deposit(300);
        b1.display();
        System.out.println();
```

```
C:\Windows\System32\cmd.e

C:\Users\DAC-User32\Desktop\nakul>javac Bank.java

C:\Users\DAC-User32\Desktop\nakul>java Bank
Interest rate is: 5.52

account number:123
account type:saving
balance100.0
Interest Earned 5.52

account number:123
account type:saving
balance400.0
Interest Earned 22.08

account number:123
account type:saving
balance300.0
Interest Earned 16.56


C:\Users\DAC-User32\Desktop\nakul>
```

7. Using this in Constructors: Create a class Rectangle with attributes length and width. Implement a parameterized constructor that initializes these attributes. Use this to differentiate between class variables and constructor parameters. Include methods to calculate the area and perimeter.

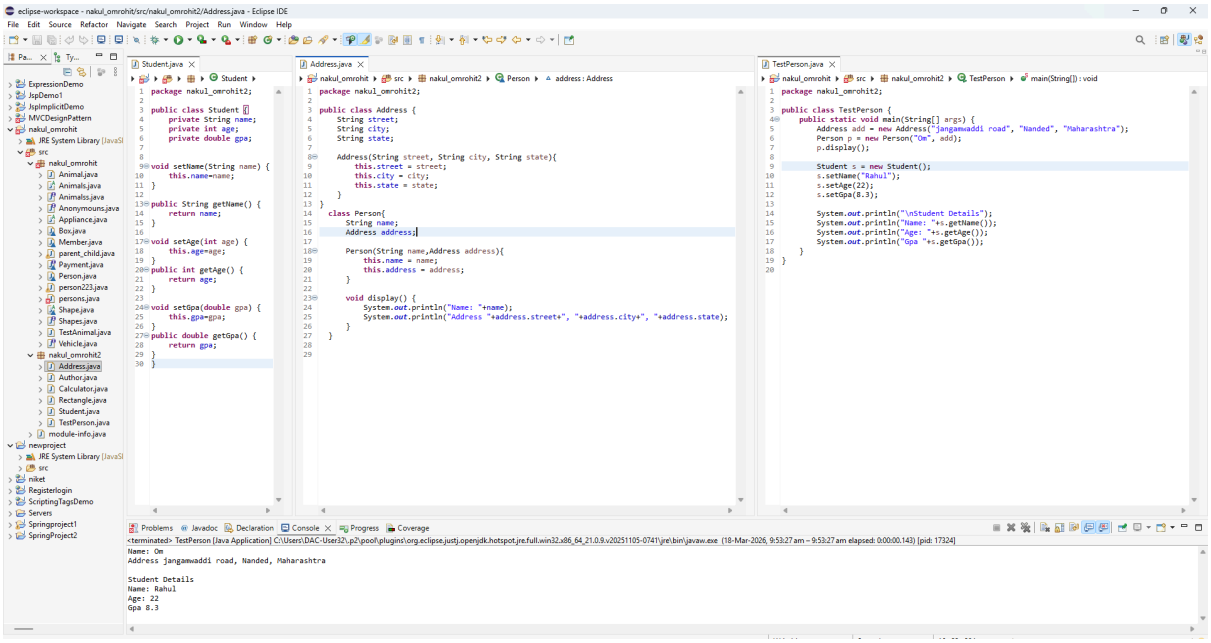
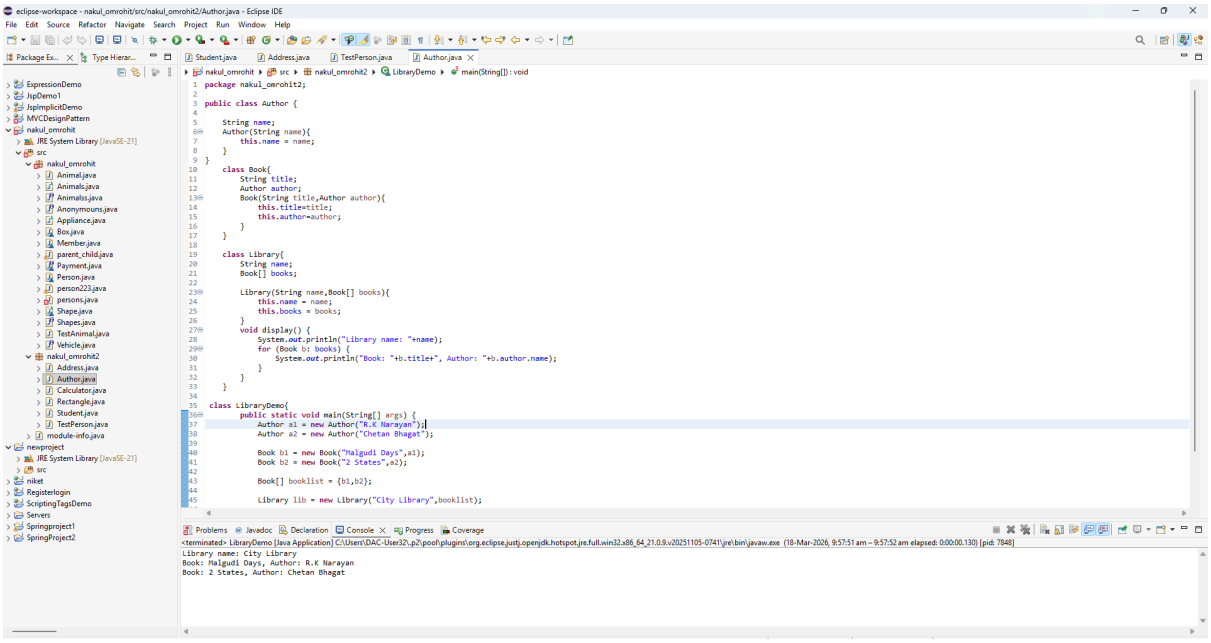


8. Class and methods: Create a class Calculator with relevant data members and a constructor. Implement methods for basic arithmetic operations (addition, subtraction, multiplication, division, modulus) and demonstrate their usage.



9. Composition and Aggregation: Create a class Address with attributes street, city, and state. Then create a class Person with attributes name and an Address object. Demonstrate how to use com Write a Java class representing a Student. Encapsulate the student's name, age, and grade point average (GPA) with private access modifiers. Provide getter and setter methods to access

and modify these attributes position to model the relationship between a person and their address



10. Implement a Java program that models a Library. Create classes for Library, Book, and Author. Ensure that the Library class aggregates a collection of Book objects, and each Book object has an aggregation relationship with an Author object.